LESSON PLAN

AFCS (22)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	07
Total No. of Lesson Plans	10	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Radar homing and sources of error and low altitude limitations
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Radar homing and sources of error and low altitude limitations.
5. Specific Objectives	22.1.1.1Trainee will be able to understand the basic of guided weapon systems and types of targets.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of radar homing principles and error analysis using IB and L & Ds; Instructor supervision required for safety protocols regarding weapon system theory.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a radar homing system?
(b) List two common types of targets for guided weapons.
(c) Define the term 'line of sight' in the context of radar guidance.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Basic principles of radar homing guidance • Types of guided weapon systems • Classification of target types • Radar signal transmission and reception • Lock-on mechanisms and engagement	L & Ds	PPT, IB	15 min
(b)	• Sources of error in radar homing • Atmospheric interference effects • Clutter and ground reflection issues • Electronic counter-measures impact • Calibration and alignment errors	L & Ds	PPT, PDF	20 min
(c)	• Low altitude flight limitations • Terrain masking and shadowing • Ground clutter dominance • Signal attenuation near surface • Navigation constraints at low levels	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how atmospheric conditions can introduce errors in radar homing.
(b) Describe the specific challenges faced during low-altitude radar operations.
(c) Identify three sources of error that affect radar homing accuracy.

12. Home Assignment and Follow Up	Prepare a brief report comparing inertial navigation and radar homing systems, focusing on their respective advantages and limitations for future study in LP 08.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Long range guidance: Inertial navigation, Hybrid inertial system and air flying missile guidance.	Refer to the master syllabus reference list	LP 08

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min